

Practical week 1

Session 1:

Task 1:

```
jella_pragathi@LAPTOP-RQQ... x + v
GNU nano 8.7.1 command_execution.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }

    else if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, NULL);

        printf("Command execution failed\n");
        exit(1);
    }

    else
    {
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
```

```
        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```
jella_pragathi@LAPTOP-RQQGT555:~$ nano command_execution.c
jella_pragathi@LAPTOP-RQQGT555:~$ gcc command_execution.c -o command_execution
jella_pragathi@LAPTOP-RQQGT555:~$ ./command_execution
Enter a Linux command: ls

Parent Process
Parent PID: 780

Child Process
Child PID: 781
Parent PID: 780
File1.txt  command_execution  parser      task1.c  task4      tokenizer.c
File2.txt  command_execution.c parser.c     task2    task4.c
OSSP       history            process     task2.c  task5
buffer     history.c          process.c   task3    task5.c
buffer.c   ossp_project       task1      task3.c  tokenizer
Child process completed.
jella_pragathi@LAPTOP-RQQGT555:~$ |
```

Task 2:

```
jella_pragathi@LAPTOP-RQQGT555:~$ child_process_completed
jella_pragathi@LAPTOP-RQQGT555:~$ uname
Linux
jella_pragathi@LAPTOP-RQQGT555:~$ uname -a
Linux LAPTOP-RQQGT555 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
jella_pragathi@LAPTOP-RQQGT555:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               46 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      18
  On-line CPU(s) list:       0-17
Vendor ID:                   GenuineIntel
Model name:                  Intel(R) Core(TM) Ultra 5 125H
CPU family:                  6
Model:                      170
Thread(s) per core:         2
Core(s) per socket:         9
Socket(s):                   1
Stepping:                    4
BogoMIPS:                    5990.39
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bml avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt fls
                             sh_lld arch_capabilities

Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
```

```
jella_pragathi@LAPTOP-RQQ... X + v
Caches (sum of all):
  L1d: 432 KiB (9 instances)
  L1i: 576 KiB (9 instances)
  L2: 18 MiB (9 instances)
  L3: 18 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-17
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Not affected
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected
jella_pragathi@LAPTOP-RQQGT555:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
/
jella_pragathi@LAPTOP-RQQGT555:~$ ps
PID TTY TIME CMD
```

```
jella_pragathi@LAPTOP-RQQ... X + v
648 pts/0 00:00:00 ps
jella_pragathi@LAPTOP-RQQGT555:~$ ps -ef
UID PID PPID C STIME TTY TIME CMD
root 1 0 0 06:36 ? 00:00:00 /sbin/init
root 2 1 0 06:36 hvc0 00:00:00 /init
root 7 2 0 06:36 hvc0 00:00:00 plan9 --control-socket 7
root 50 1 0 06:36 ? 00:00:00 /usr/lib/systemd/systemd
systemd+ 82 1 0 06:36 ? 00:00:00 /usr/lib/systemd/systemd
root 90 1 0 06:36 ? 00:00:00 /usr/lib/systemd/systemd
root 136 1 0 06:36 ? 00:00:00 /bin/sh /usr/lib/systemd
root 144 1 0 06:36 ? 00:00:00 /usr/sbin/cron -f -P
message+ 150 1 0 06:36 ? 00:00:00 @dbus-daemon --system --
root 168 1 0 06:36 ? 00:00:00 /usr/bin/python3 /usr/bi
root 173 1 0 06:36 ? 00:00:00 /usr/lib/systemd/systemd
root 176 1 0 06:36 ? 00:00:00 /usr/libexec/wsl-pro-ser
syslog 190 1 0 06:36 ? 00:00:00 /usr/sbin/rsyslogd -n -i
root 249 1 0 06:36 ? 00:00:00 /usr/bin/python3 /usr/sh
_chrony 253 136 0 06:36 ? 00:00:00 /usr/sbin/chronyd -n -F
_chrony 269 253 0 06:36 ? 00:00:00 /usr/sbin/chronyd -n -F
root 292 1 0 06:36 tty1 00:00:00 /usr/sbin/agetty --nores
root 326 2 0 06:36 ? 00:00:00 /init
root 327 326 0 06:36 ? 00:00:00 /init
jella_pt 331 327 0 06:36 pts/0 00:00:00 -bash
root 332 2 0 06:36 ? 00:00:00 login -- jella_pragathi
jella_pt 393 1 0 06:36 ? 00:00:00 /usr/lib/systemd/systemd
jella_pt 395 393 0 06:36 ? 00:00:00 (sd-pam)
jella_pt 424 332 0 06:36 pts/1 00:00:00 -bash
jella_pt 660 331 0 06:43 pts/0 00:00:00 ps -ef
```

```
jella_pragathi@LAPTOP-RQQGT555:~$ top
top - 06:48:35 up 12 min, 1 user, load average: 0.04, 0.07, 0.01
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7651.2 total, 7032.9 free, 605.9 used, 163.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7045.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
82	systemd+	20	0	22416	14432	11940	S	0.3	0.2	0:00.10	systemd-resolve
1	root	20	0	24200	15428	11608	S	0.0	0.2	0:00.93	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3180	2048	1940	S	0.0	0.0	0:00.00	init
50	root	19	-1	50404	16812	15544	S	0.0	0.2	0:00.25	systemd-journal
90	root	20	0	35452	12360	9188	S	0.0	0.2	0:00.57	systemd-udev
136	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.09	chronyd-starter
144	root	20	0	4472	3128	2884	S	0.0	0.0	0:00.01	cron
150	message+	20	0	8796	5356	4676	S	0.0	0.1	0:00.07	dbus-daemon
168	root	20	0	44096	29728	14580	S	0.0	0.4	0:00.33	networkd-dispat
173	root	20	0	18440	9412	8168	S	0.0	0.1	0:00.10	systemd-logind
190	syslog	20	0	220548	5908	4616	S	0.0	0.1	0:00.09	rsyslogd
249	root	20	0	123460	32552	17940	S	0.0	0.4	0:00.22	unattended-upgr
253	_chrony	20	0	23740	11196	7204	S	0.0	0.1	0:00.08	chronyd
269	_chrony	20	0	12096	2472	1824	S	0.0	0.0	0:00.00	chronyd
292	root	20	0	5492	2720	2536	S	0.0	0.0	0:00.01	agetty
326	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
327	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.05	Relay(331)
331	jella_p+	20	0	6268	5544	3864	S	0.0	0.1	0:00.11	bash
332	root	20	0	8648	5620	4776	S	0.0	0.1	0:00.01	login
393	jella_p+	20	0	22344	13560	10928	S	0.0	0.2	0:00.16	systemd
395	jella_p+	20	0	22920	4072	2072	S	0.0	0.1	0:00.00	(sd-pam)
424	jella_p+	20	0	6136	5488	3928	S	0.0	0.1	0:00.04	bash
672	jella_p+	20	0	8172	6052	3936	R	0.0	0.1	0:00.15	top